Git & GitHub Fundamentals

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1. Final Repository & Ignore State

The screenshot below demonstrates the final state of the remote repository on GitHub. It confirms the successful push of the `main` branch, including the `.gitignore` file, the edited `README.md`, the tracked `about.txt` from the merged feature branch, and the uploaded image asset.

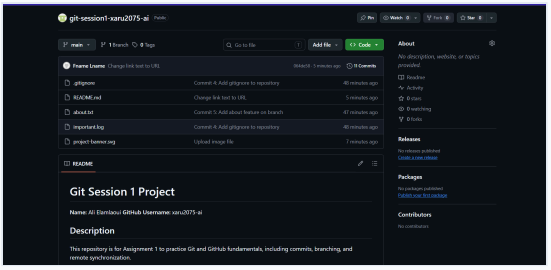


Figure 1: GitHub repository file listing showing 2 branches and 11 total commits.

2. Rendered README

This section captures the properly formatted `README.md` as rendered by GitHub. It validates the correct use of Markdown syntax for the Python code block, the embedded hyperlink, and the displayed SVG banner image.

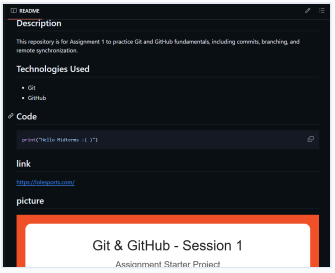


Figure 2: Formatted README showing the code block, link, and banner image.

3. Compact Commit & Branch/Merge History

Using the `git log --oneline --graph --all` command, the terminal output clearly visualizes the commit history. It exceeds the 5-commit requirement (showing 11 total commits) and successfully proves the creation, progression, and merging of the `feature-about` branch into the `main` branch.

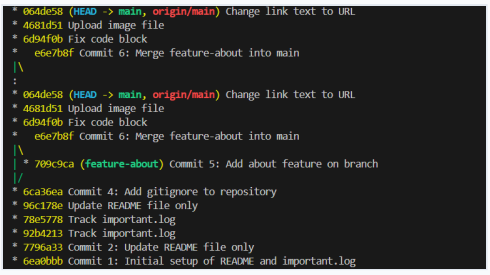


Figure 3: Compact visual graph of the branching and merging history.

4. Selective Staging

This state demonstrates selective staging using `git status`. The `README.md` file was explicitly added to the staging area (indicated in green), while modifications to `important.log` were intentionally left unstaged (indicated in red).

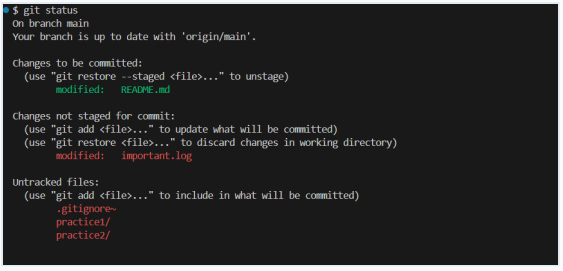


Figure 4: Terminal output verifying selective file staging.

5. Local Main Behind Origin/Main

To satisfy the remote tracking requirement, a test commit was made directly on GitHub. After executing `git fetch`, the local status correctly identifies that the local `main` branch is behind `origin/main` by 1 commit and can be fast-forwarded.

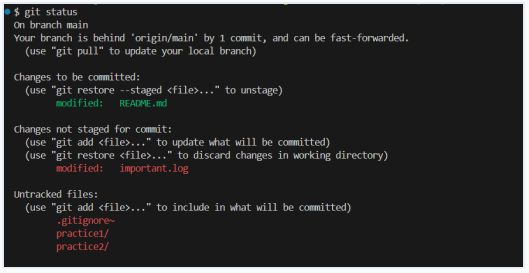


Figure 5: Status indicating the local branch is behind the remote origin.

6. Workflow Notes

I occasionally consulted an AI assistant during this assignment just to double-check some minor Markdown formatting syntax and quickly reference a couple of basic terminal commands. The repository configuration, branching logic, selective staging, and overall execution of the assignment were handled manually.